



Network Management and Administration (IE2110)
2nd Year, 2nd Semester

Assignment

Zimbra

Submitted to
Sri Lanka Institute of Information Technology

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In partial fulfillment of the requirements for the
Bachelor of Science Special Honors Degree in Information Technology

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Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

Introduction

This assignment required the deployment and validation of a Linux-based network environment capable of delivering several core administrative services within a controlled virtual laboratory. The completed environment covers the full set of required functions: initial virtual machine preparation, network configuration, dynamic IP address assignment, electronic mail services, centralized directory and authentication services, and web publishing.

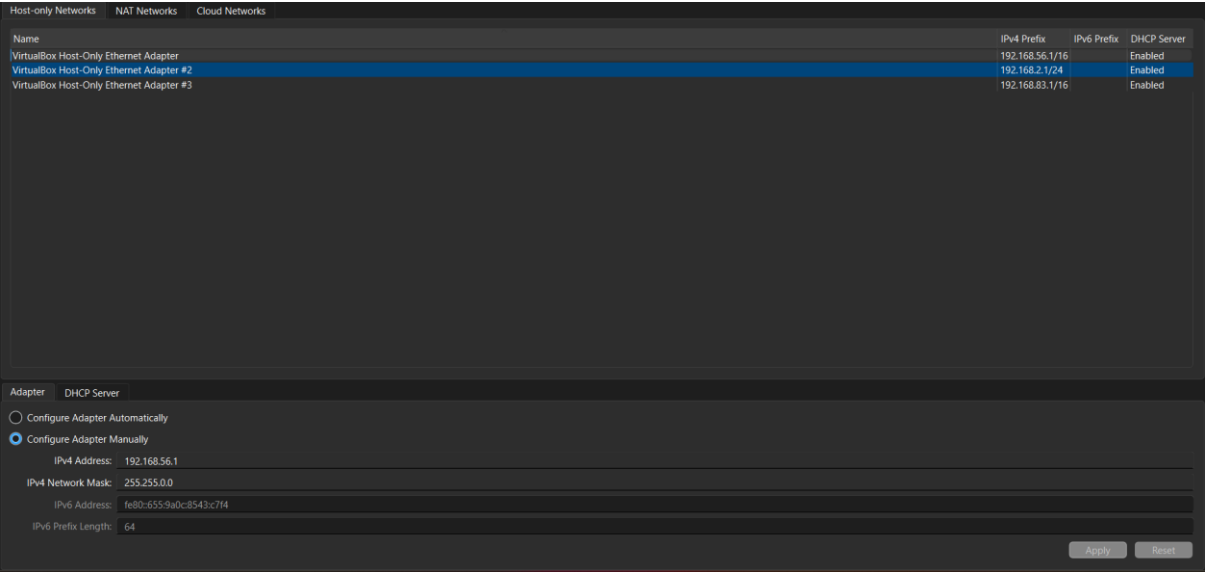
The implementation used the assignment-specific values of network **192.168.83.0/24**, DHCP range **192.168.83.100 - 192.168.83.200**, mail domain **mail.it83.local**, and Active Directory domain **it83.local**.

The assignment required Zimbra and Samba to be installed and configured on Ubuntu Server (22.04), while the web server was implemented using CentOS 10.

After the configuration phase, the environment was tested to verify that all major services were functioning correctly. The validation confirmed successful DHCP lease allocation, proper mail account creation and email communication through Zimbra, correct domain and user management through Samba Active Directory, successful domain access from the Ubuntu client machine, and proper client-side access to the hosted web page. Therefore, the completed system demonstrates a fully operational Linux-based network environment.

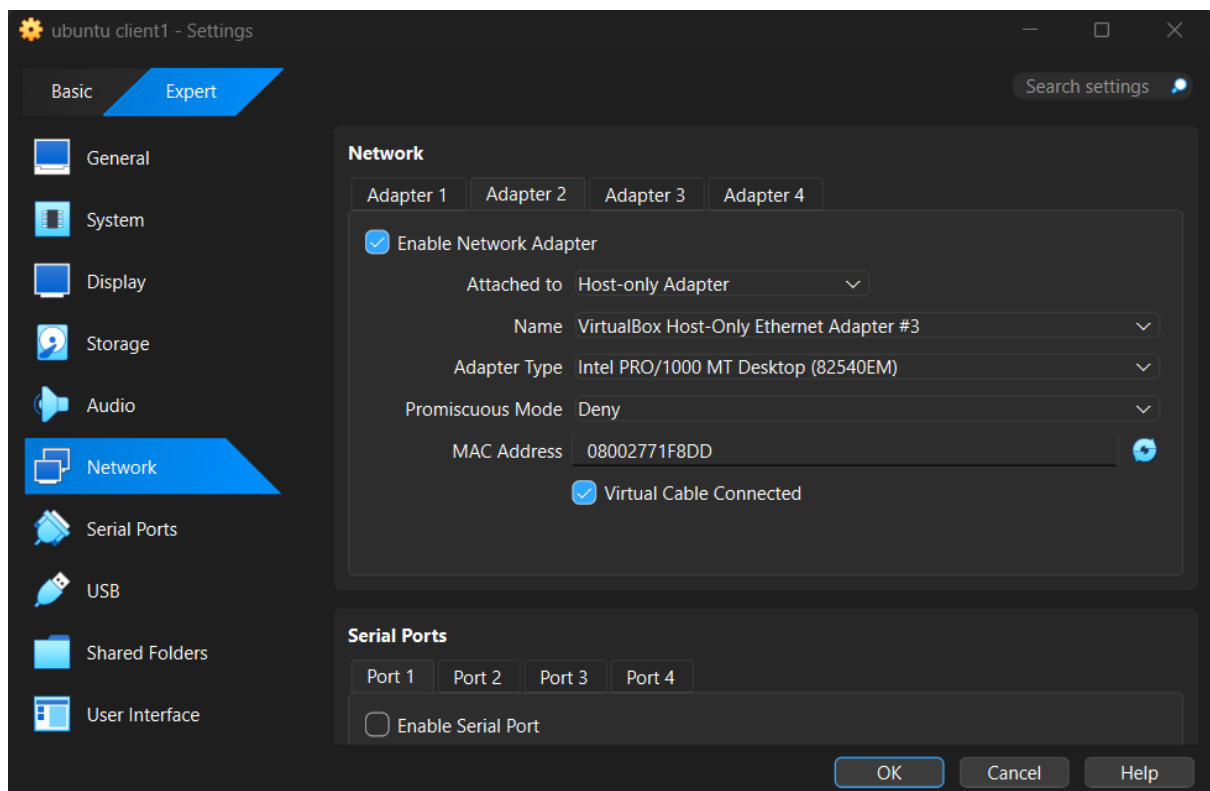
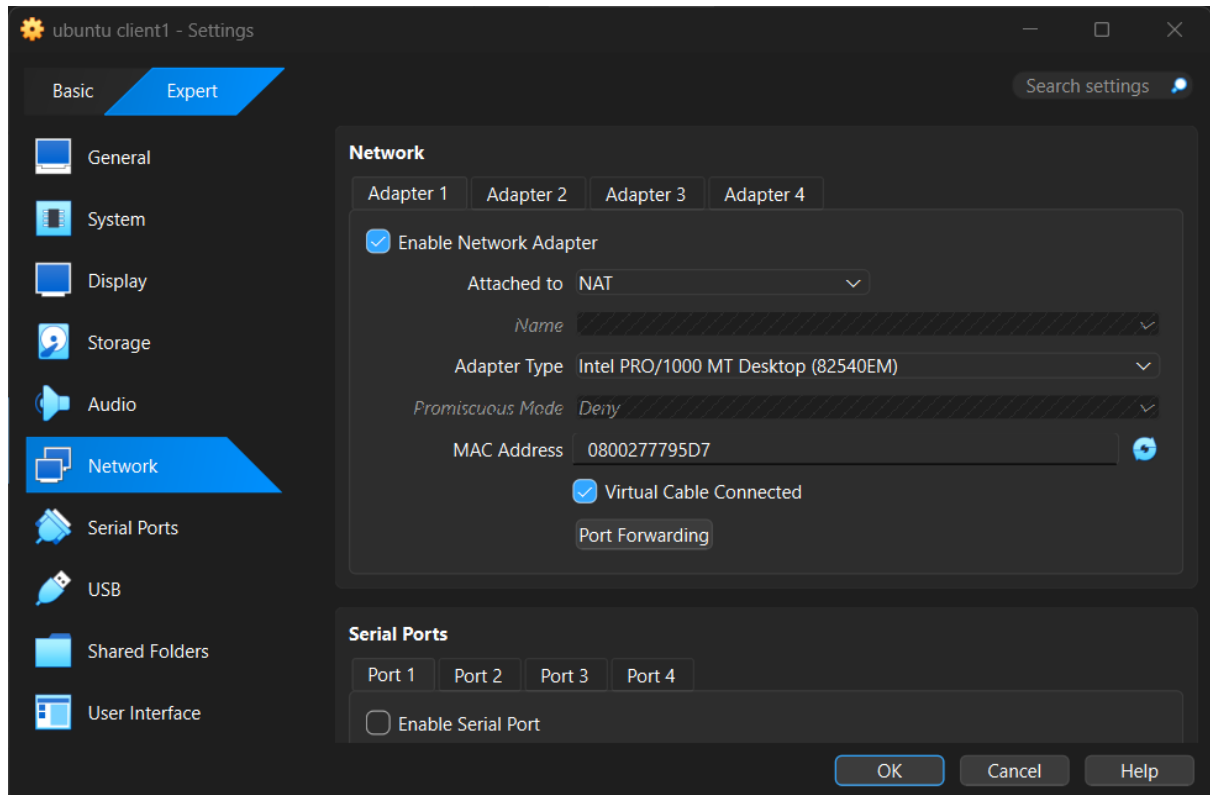
Part A: Initial Setup

Virtual network preparation

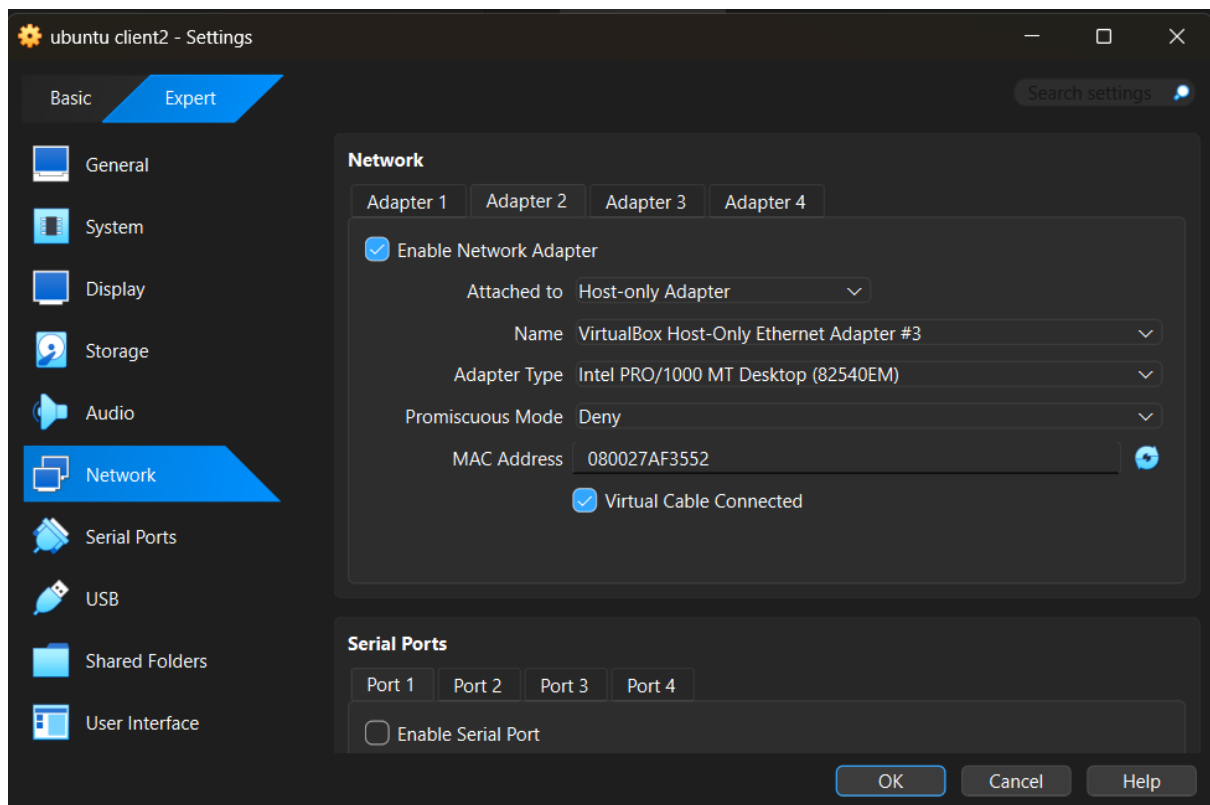
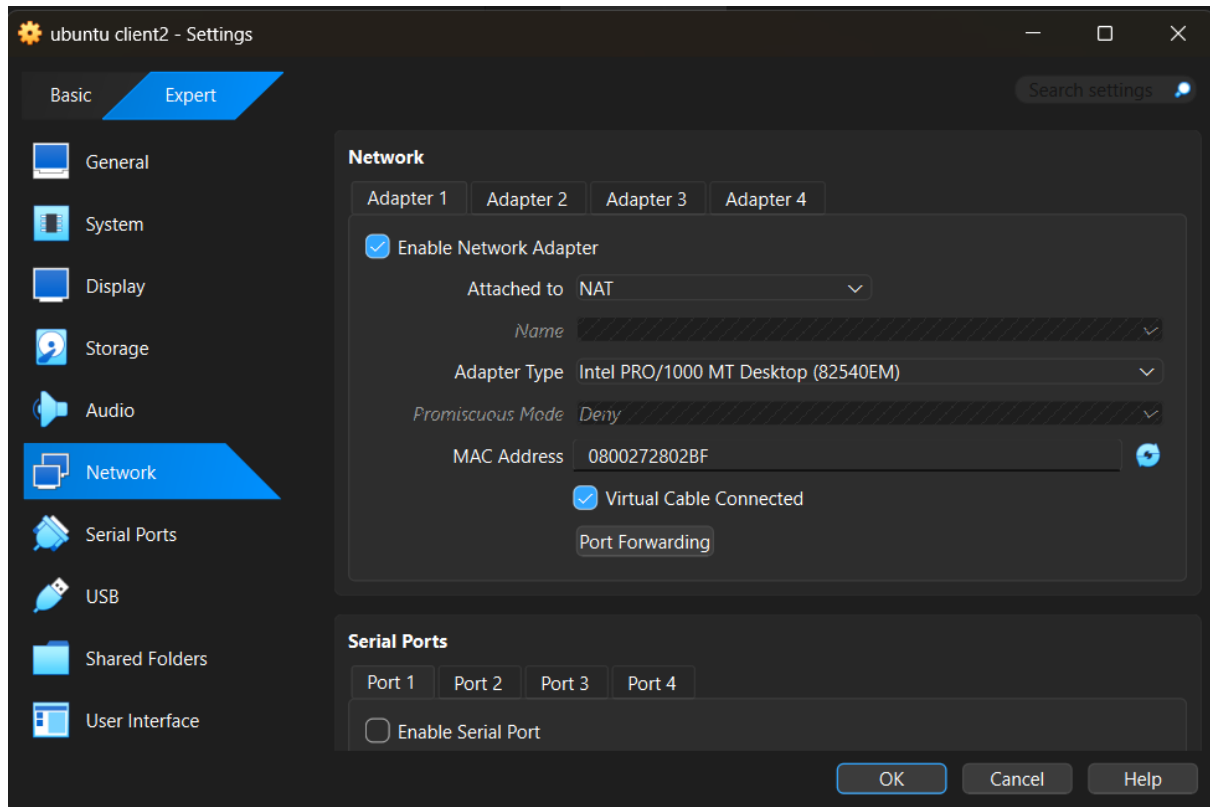


The following diagram illustrates the VirtualBox host-only network configuration established for the internal lab environment. In this setup, two Ubuntu servers have been configured to provide Zimbra mail services and Samba Active Directory services respectively, while a CentOS server is used as the web server. Due to an issue encountered with Adapter 2, an additional network adapter (Adapter 3) was introduced and configured as the host-only interface. Adapter 3 has been assigned the IP address 192.168.83.1, which serves as the base address of the internal lab network.

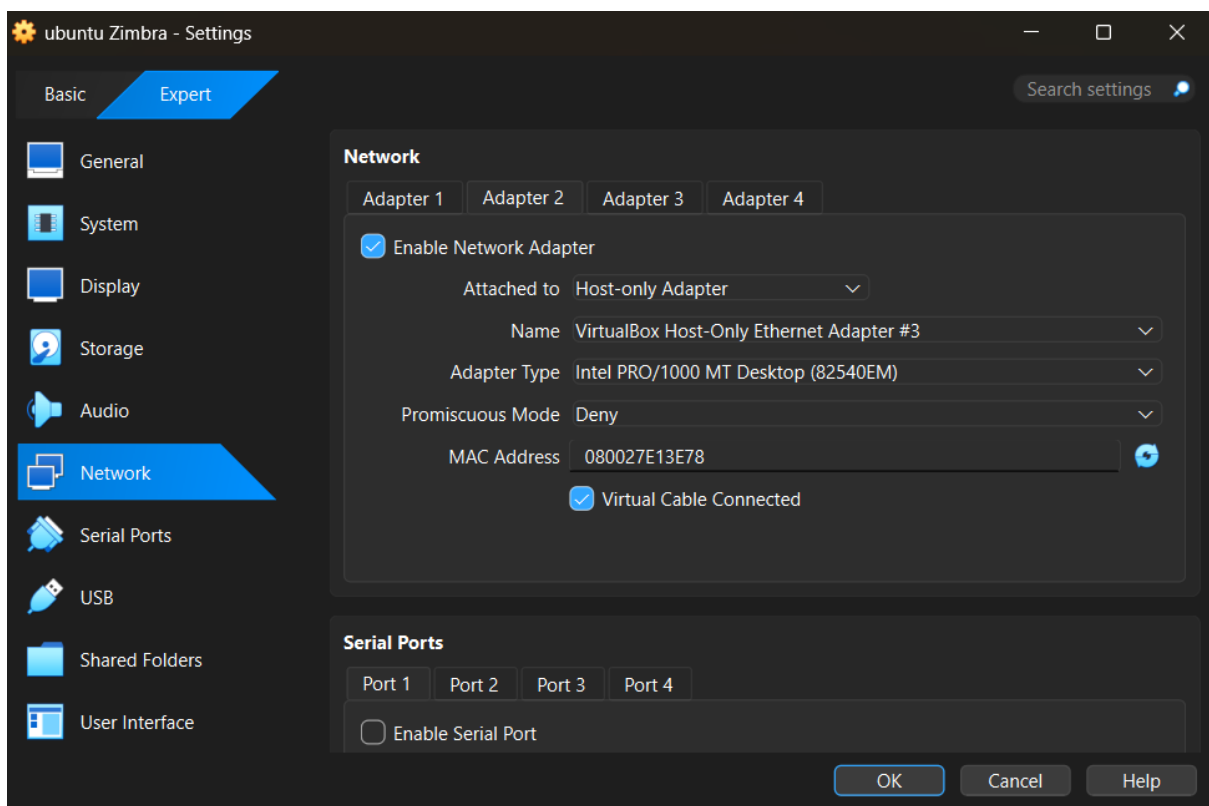
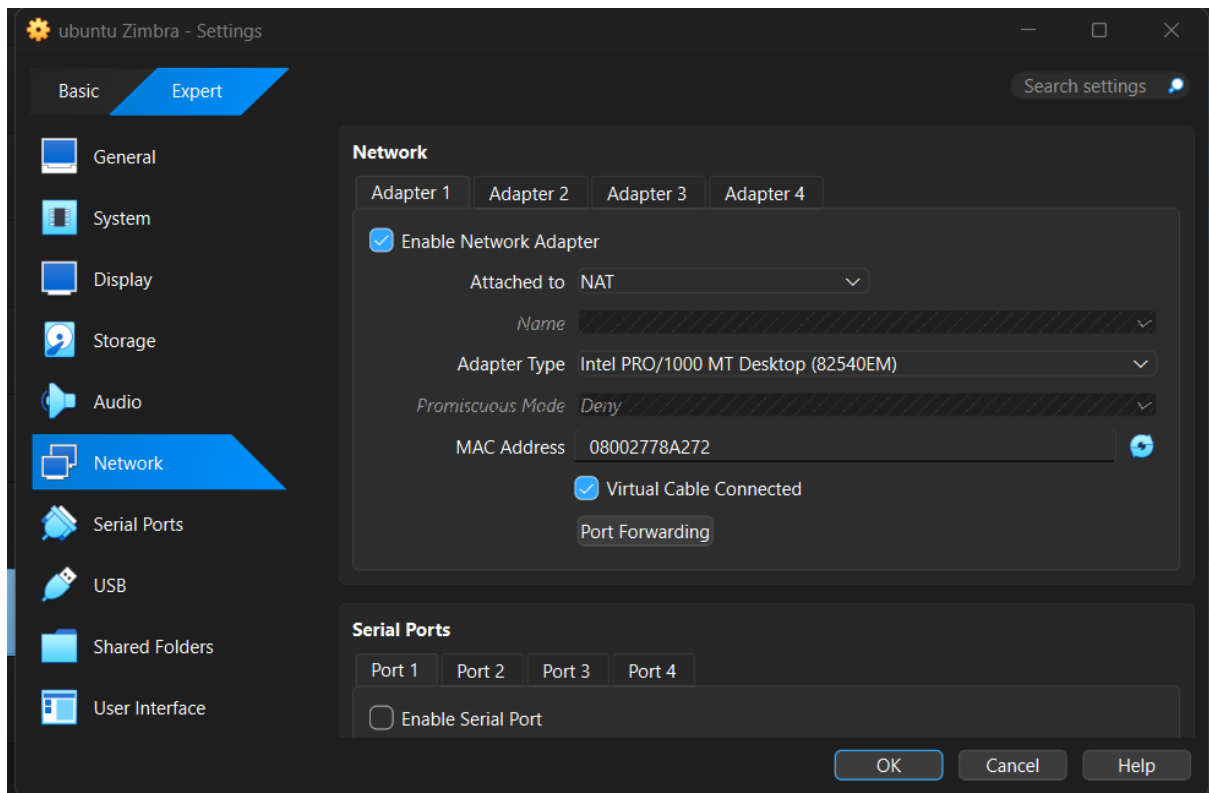
Ubuntu client 01



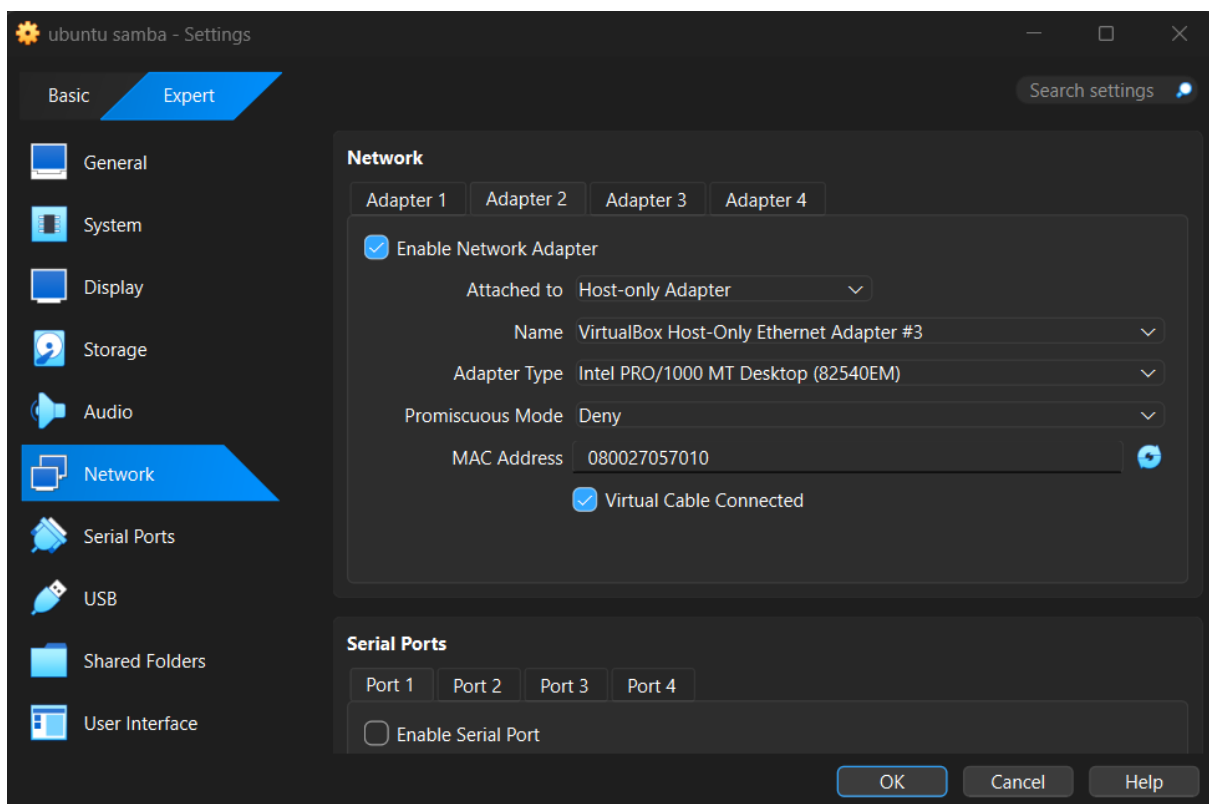
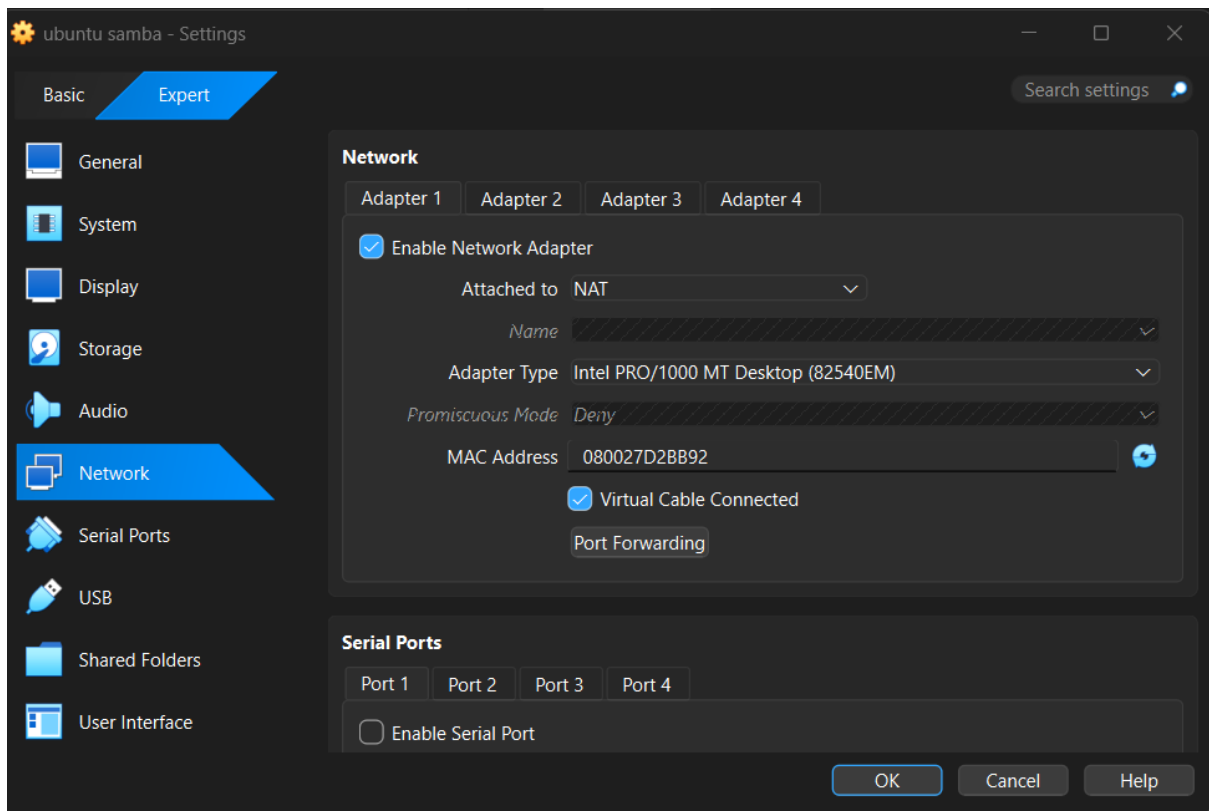
Ubuntu client 2



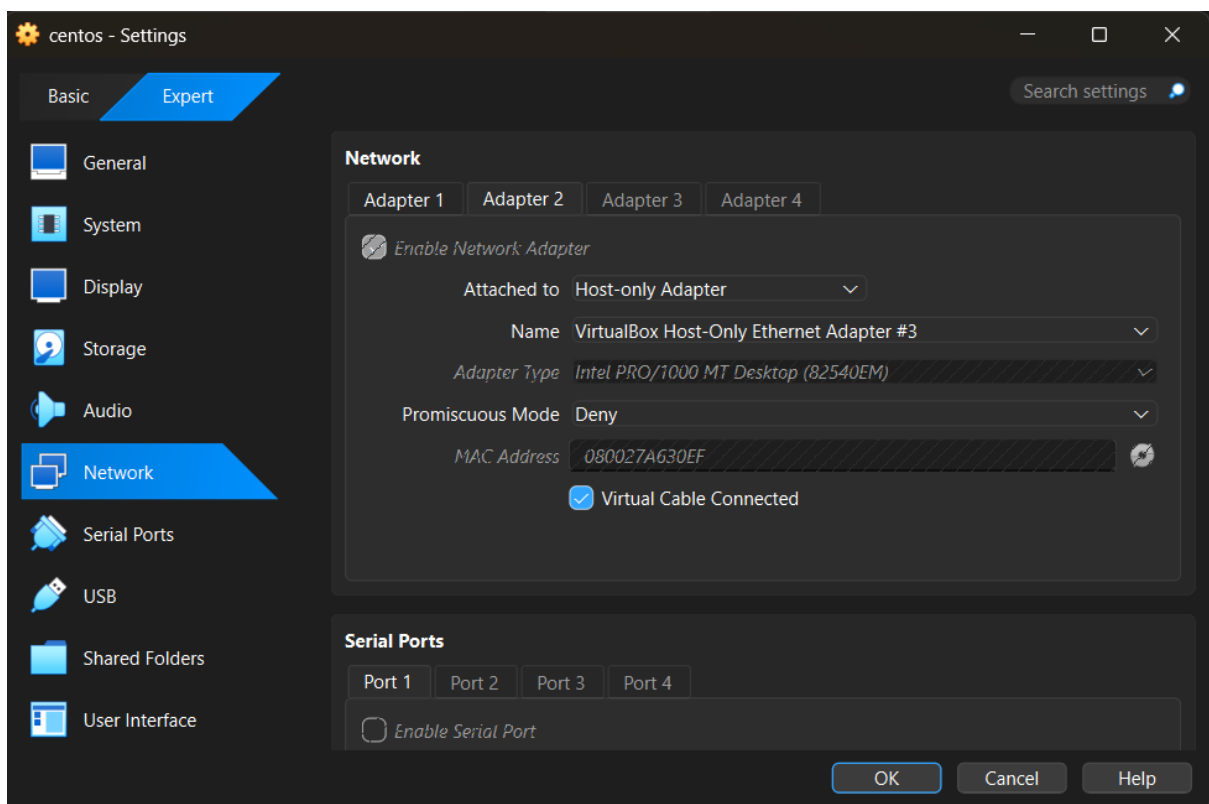
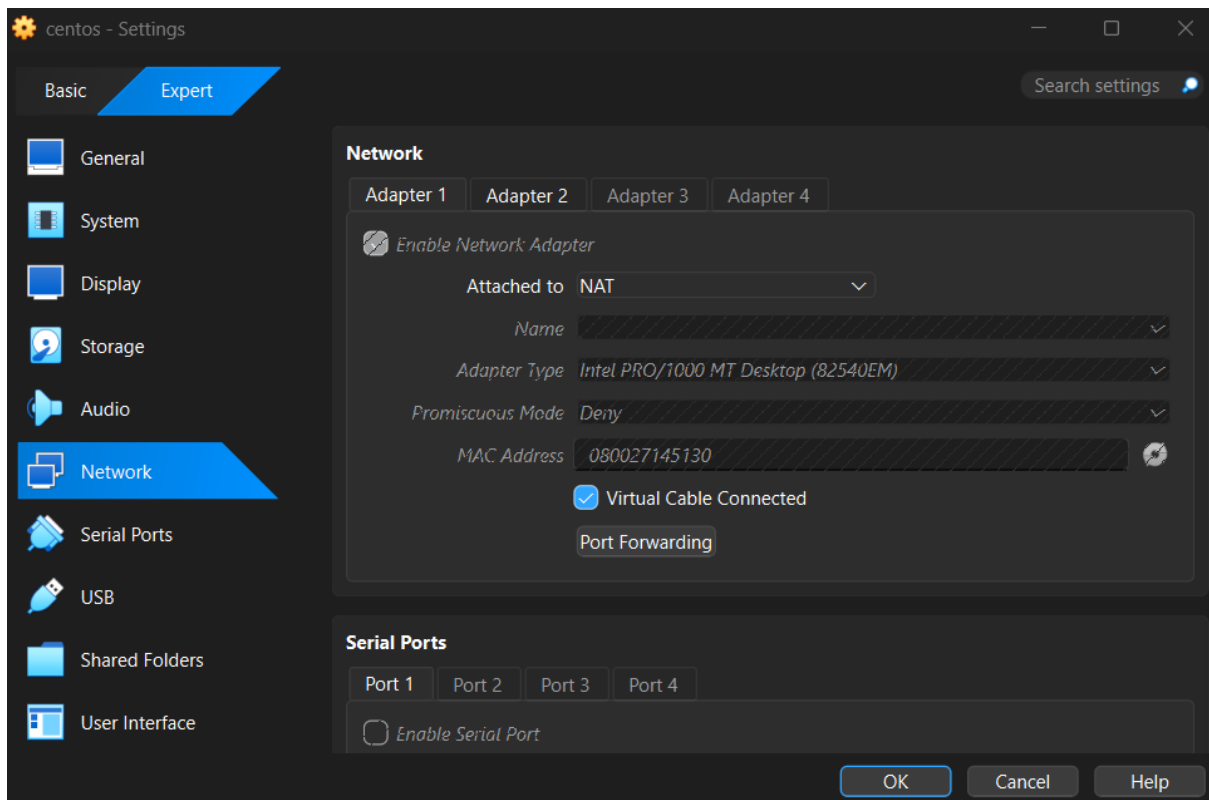
Zimbra server (Ubuntu 22.04)



Samba Active Directory Domain Controller (Ubuntu Server 22.04)



Web Server(Centos 10)



For the two client virtual machines, Adapter 1 was configured in NAT mode to allow internet connectivity for software installation, updates, and general external access. Adapter 3 was configured as Host-Only so that both clients could communicate within the internal laboratory network. For the server machines, Adapter 1 was set to Bridged mode to enable direct network connectivity and support SSH access from the host or other devices on the physical network. In addition, Adapter 3 was configured as Host-Only to connect the servers to the isolated internal lab segment used for service testing and client-server communication.

Interface and connectivity validation

Ubuntu (Zimbra)

```
Last login: Sun Apr 12 15:32:16 UTC 2026 from 192.168.83.1 on pts/0
sasmi@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:18:25:9d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86357sec preferred_lft 86357sec
    inet6 fd17:625c:f037:2:a00:27ff:fe18:259d/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86360sec preferred_lft 14360sec
    inet6 fe80::a00:27ff:fe18:259d/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:27:79:e3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.83.1/24 brd 192.168.83.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe27:79e3/64 scope link
        valid_lft forever preferred_lft forever
```

Centos (web Server)

```
[root@localhost sasmil# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:14:51:30 brd ff:ff:ff:ff:ff:ff
    altname enx080027145130
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86367sec preferred_lft 86367sec
    inet6 fd17:625c:f037:2:a00:27ff:fe14:5130/64 scope global dynamic noprefixroute
        valid_lft 86369sec preferred_lft 14369sec
    inet6 fe80::a00:27ff:fe14:5130/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a6:30:ef brd ff:ff:ff:ff:ff:ff
    altname enx080027a630ef
    inet 192.168.83.100/24 brd 192.168.83.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::8d2e:8592:1c30:d910/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@localhost sasmil# _
```

Part B – DHCP Server

```
sasmi@ubuntu: ~
addr noprefixroute
    valid_lft 86272sec preferred_lft 14272sec
    inet6 fe80::a00:27ff:fe78:a272/64 scope link
    valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state U
P group default qlen 1000
    link/ether 08:00:27:e1:3e:78 brd ff:ff:ff:ff:ff:ff
    inet 192.168.83.100/24 brd 192.168.83.255 scope global dynamic enp0s8
        valid_lft 513sec preferred_lft 513sec
    inet6 fe80::a00:27ff:fe78:a272/64 scope link
        valid_lft forever preferred_lft forever
sasmi@ubuntu:~$ sudo nano /etc/default/isc-dhcp-server

sasmi@ubuntu:~$ sudo systemctl restart isc-dhcp-server
sudo systemctl enable isc-dhcp-server
Synchronizing state of isc-dhcp-server.service with SysV service script with
/lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable isc-dhcp-server
sasmi@ubuntu:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/lib/systemd/system/isc-dhcp-server.service; enabled; >
   Active: active (running) since Sun 2026-04-12 11:40:28 UTC; 9s ago
     Docs: man:dhcpcd(8)
  Main PID: 1752 (dhcpcd)
    Tasks: 4 (limit: 4554)
   Memory: 4.5M
      CPU: 18ms
   CGroup: /system.slice/isc-dhcp-server.service
           └─1752 dhcpcd -user dhcpcd -group dhcpcd -f -4 -pf /run/dhcp-serv>

Apr 12 11:40:28 ubuntu sh[1752]: PID file: /run/dhcp-server/dhcpcd.pid
Apr 12 11:40:28 ubuntu dhcpcd[1752]: Wrote 0 leases to leases file.
Apr 12 11:40:28 ubuntu sh[1752]: Wrote 0 leases to leases file.
Apr 12 11:40:28 ubuntu dhcpcd[1752]: Listening on LPF/enp0s8/08:00:27:e1:3e:78/>
Apr 12 11:40:28 ubuntu sh[1752]: Listening on LPF/enp0s8/08:00:27:e1:3e:78/>
Apr 12 11:40:28 ubuntu sh[1752]: Sending on LPF/enp0s8/08:00:27:e1:3e:78/>
Apr 12 11:40:28 ubuntu sh[1752]: Sending on Socket/fallback/fallback-net
Apr 12 11:40:28 ubuntu dhcpcd[1752]: Sending on LPF/enp0s8/08:00:27:e1:3e:78/>
Apr 12 11:40:28 ubuntu dhcpcd[1752]: Sending on Socket/fallback/fallback-n>
Apr 12 11:40:28 ubuntu dhcpcd[1752]: Server starting service.
lines 1-21/21 (END)
```

I installed kea DHCP configuration on ubuntu server for the assignment subnet.

Subnet – 192.168.83.0

Range – 192.168.83.100 – 192.168.83.200

This figure records the operational state of the Kea DHCP service after activation. It verifies that the DHCP daemon started successfully on the Ubuntu server and was available to answer lease requests from client systems. This is important because a correct configuration file alone does not satisfy the assignment; the service must also be actively running.

Part C – Zimbra Mail server

DNS installation and active

```
. bind9.service
sasmi@ubuntu:~$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor pre>
   Active: active (running) since Sun 2026-04-12 16:33:30 UTC; 23s ago
     Docs: man:named(8)
  Process: 2649 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0>
 Main PID: 2650 (named)
    Tasks: 6 (limit: 4554)
   Memory: 23.0M
      CPU: 146ms
   CGroup: /system.slice/named.service
           └─2650 /usr/sbin/named -u bind

Apr 12 16:33:30 ubuntu named[2650]: zone localhost/IN: loaded serial 2
Apr 12 16:33:30 ubuntu named[2650]: zone 127.in-addr.arpa/IN: loaded serial>
Apr 12 16:33:30 ubuntu named[2650]: zone 83.168.192.in-addr.arpa/IN: loaded>
Apr 12 16:33:30 ubuntu named[2650]: zone 255.in-addr.arpa/IN: loaded serial>
Apr 12 16:33:30 ubuntu named[2650]: zone it83.local/IN: loaded serial 2
Apr 12 16:33:30 ubuntu named[2650]: all zones loaded
Apr 12 16:33:30 ubuntu named[2650]: running
Apr 12 16:33:30 ubuntu systemd[1]: Started BIND Domain Name Server.
Apr 12 16:33:30 ubuntu named[2650]: managed-keys-zone: Key 20326 for zone .>
Apr 12 16:33:30 ubuntu named[2650]: managed-keys-zone: Key 38696 for zone .>
```

Zimbra Installation

```
zimbra-proxy-components  
...done
```

Installing local packages (26):

```
zimbra-timezone-data  
zimbra-common-core-libs  
zimbra-common-mbox-conf-attrs  
zimbra-common-mbox-db  
zimbra-common-mbox-native-lib  
zimbra-common-mbox-conf  
zimbra-common-mbox-conf-msgs  
zimbra-common-core-jar  
zimbra-common-mbox-conf-rights  
zimbra-common-mbox-docs  
zimbra-core  
zimbra-ldap  
zimbra-logger  
zimbra-mta  
zimbra-dnscache  
zimbra-snmp  
zimbra-mbox-conf  
zimbra-mbox-admin-console-war  
zimbra-mbox-war  
zimbra-mbox-service  
zimbra-mbox-webclient-war  
zimbra-mbox-store-libs  
zimbra-store  
zimbra-apache  
zimbra-spell  
zimbra-proxy  
...done
```

```
Installing common zimlets...
  com_zimbra_mailarchive...done.
  com_zimbra_email...done.
  com_zimbra_phone...done.
  com_zimbra_srchhighlighter...done.
  com_zimbra_proxy_config...done.
  com_zimbra_attachmail...done.
  com_zimbra_cert_manager...done.
  com_zimbra_date...done.
  com_zimbra_webex...done.
  com_zimbra_viewmail...done.
  com_zimbra_tooltip...done.
  com_zimbra_ymemoticons...done.
  com_zimbra_url...done.
  com_zimbra_adminversioncheck...done.
  com_zimbra_attachcontacts...done.
  com_zimbra_gotourl...done.
  com_zimbra_bulkprovision...done.
Finished installing common zimlets.
Getting list of all zimlets...done.
Updating non-standard zimlets...
Finished updating non-standard zimlets.
Restarting mailboxd...done.
Skipping creation of default domain GAL sync account - existing install detected.

You have the option of notifying Zimbra of your installation.
This helps us to track the uptake of the Zimbra Collaboration Server.
The only information that will be transmitted is:
  The VERSION of zcs installed (10.1.16_GA_4200001_UBUNTU22_64)
  The ADMIN EMAIL ADDRESS created (admin@mail.it83.local)

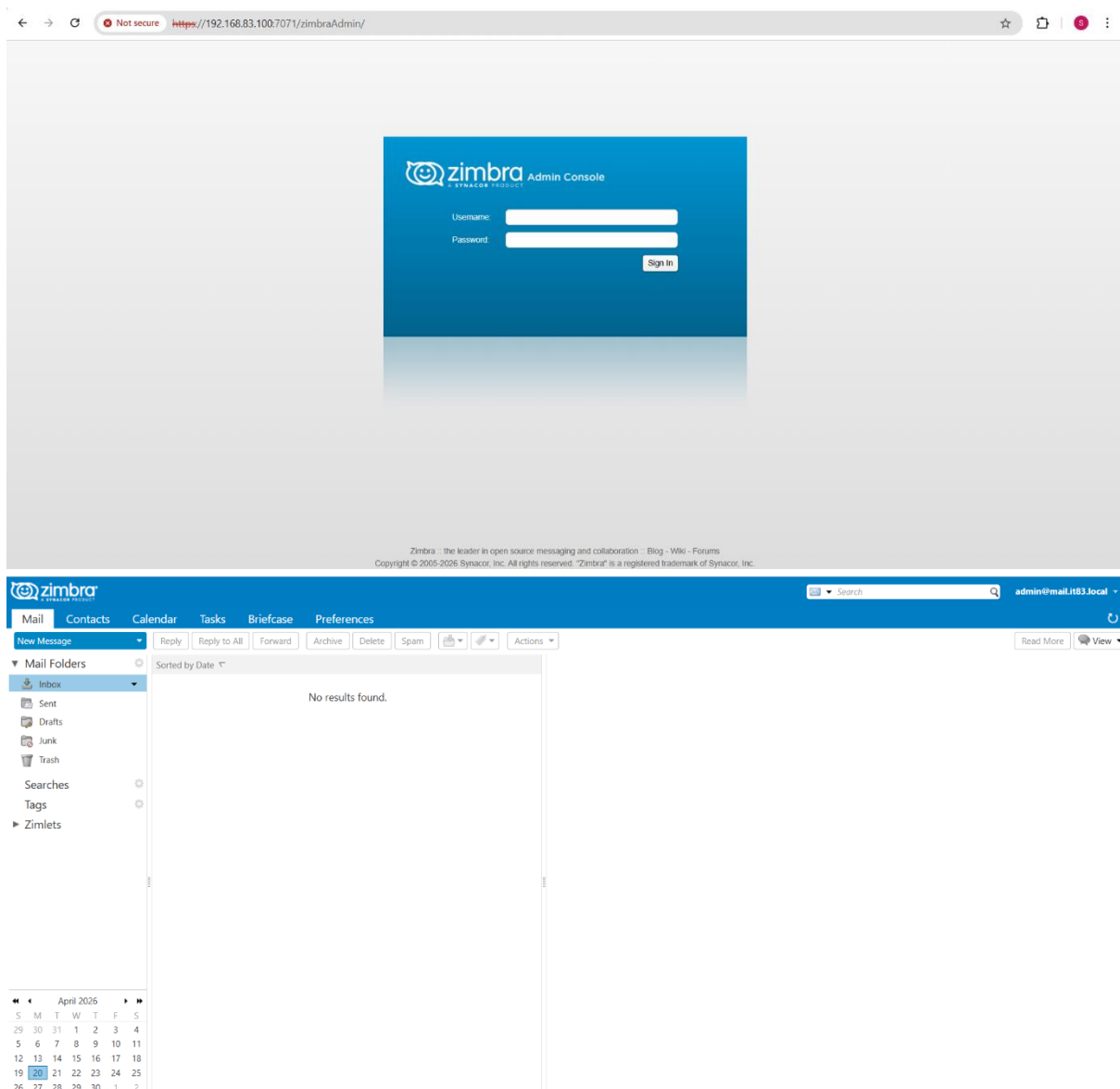
Notify Zimbra of your installation? [Yes] no
Notification skipped
Setting up zimbra crontab...done.

Moving /tmp/zmsetup.20260418-102539.log to /opt/zimbra/log

Configuration complete - press return to exit
```

The above two figures illustrate the final summary of Zimbra Collaboration Suite setup. It is evident that all necessary components of Zimbra such as LDAP, MTA, mailbox storage, proxy, and logging have been enabled successfully. The mail server name has been set as mail.it83.local, and ports for all essential services have been allocated automatically. The screen clearly indicates that the setup process has finally come to its final review step where the settings can be saved and implemented. Despite the unavailability of any license information, the necessary mail server configuration has been created successfully.

Zimbra mail proof



First figure illustrates the successful access to the Zimbra Web Client login interface via the server IP address. The second one demonstrates a successful login to the Zimbra mail system under the mail.it83.local domain, displaying the user inbox and confirming that the webmail service is properly configured and operational. This verifies that users are able to authenticate and access their email accounts through the Zimbra web interface

Part D – Samba Mail Server

Samba is an open source software package that enables Linux servers to act like a Windows network by offering file and printer sharing along with directory services. For this task, the Samba software package has been used to configure an Active Directory Domain Controller

for managing users, authenticating them and controlling access to their domains from clients in the it83.local network.

```
sasmi@dc1: ~
● samba-ad-dc.service - Samba AD Daemon
   Loaded: loaded (/lib/systemd/system/samba-ad-dc.service; enabled; vend
   Active: active (running) since Sun 2026-04-12 12:50:53 UTC; 8s ago
     Docs: man:samba(8)
           man:samba(7)
           man:smb.conf(5)
  Main PID: 5097 (samba)
    Status: "samba: ready to serve connections..."
     Tasks: 60 (limit: 4554)
    Memory: 198.3M
       CPU: 5.732s
   CGroup: /system.slice/samba-ad-dc.service
           └─5097 "samba: root process" "" "" "" "" "" "" "" "" "" "" "" >
             └─5098 "samba: tfork waiter process(5099)" "" "" "" "" "" "" "" >
               └─5099 "samba: task[s3fs] pre-fork master" "" "" "" "" "" "" "" >
                 └─5100 "samba: tfork waiter process(5101)" "" "" "" "" "" "" "" >
                   └─5101 "samba: task[rpc] pre-fork master" "" "" "" "" "" "" "" >
                     └─5102 "samba: tfork waiter process(5104)" "" "" "" "" "" "" "" >
                       └─5103 "samba: tfork waiter process(5106)" "" "" "" "" "" "" "" >
                         └─5104 /usr/sbin/smbd -D "--option=server role check:inhibit=y>
                           └─5105 "samba: tfork waiter process(5107)" "" "" "" "" "" "" "" >
                             └─5106 "samba: task[nbt] pre-fork master" "" "" "" "" "" "" "" >
                               └─5107 "samba: task[rpc] pre-forked worker(0)" "" "" "" "" "" "" "" >
                                 └─5108 "samba: tfork waiter process(5110)" "" "" "" "" "" "" "" >
                                   └─5109 "samba: tfork waiter process(5112)" "" "" "" "" "" "" "" >
                                     └─5110 "samba: task[wrepl] pre-fork master" "" "" "" "" "" "" "" >
                                       └─5111 "samba: tfork waiter process(5116)" "" "" "" "" "" "" "" >
                                         └─5112 "samba: task[rpc] pre-forked worker(1)" "" "" "" "" "" "" "" >
                                           └─5113 "samba: tfork waiter process(5114)" "" "" "" "" "" "" "" >
                                             └─5114 "samba: task[rpc] pre-forked worker(2)" "" "" "" "" "" "" "" >
                                               └─5115 "samba: tfork waiter process(5118)" "" "" "" "" "" "" "" >
                                                 └─5116 "samba: task[ldap] pre-fork master" "" "" "" "" "" "" "" >
                                                   └─5117 "samba: tfork waiter process(5119)" "" "" "" "" "" "" "" >
                                                     └─5118 "samba: task[rpc] pre-forked worker(3)" "" "" "" "" "" "" "" >
                                                       └─5119 "samba: task[cldap] pre-fork master" "" "" "" "" "" "" "" >
                                                         └─5120 "samba: tfork waiter process(5121)" "" "" "" "" "" "" "" >
                                                           └─5121 "samba: task[kdc] pre-fork master" "" "" "" "" "" "" "" >
                                                             └─5122 "samba: tfork waiter process(5123)" "" "" "" "" "" "" "" >
                                                               └─5123 "samba: task[drepl] pre-fork master" "" "" "" "" "" "" "" >
                                                                 └─5124 "samba: tfork waiter process(5126)" "" "" "" "" "" "" "" >
lines 1-40
```

It is evident from this figure that the Samba Active Directory Domain Controller was successfully launched, and the service is running in an active state on the Ubuntu server. This confirms that the system is fully prepared to handle all domain-related operations, including directory services, user authentication, and administrative tasks within the it83.local domain. Ensuring that the service is actively running is essential, as it guarantees that the domain

controller functions correctly and can effectively support authentication and management activities within the lab environment.

```
sasmi@dc1:~$ sudo samba-tool user create admin1 Password@123
User 'admin1' added successfully
sasmi@dc1:~$ sudo samba-tool user create staff1 Password@123
User 'staff1' added successfully
sasmi@dc1:~$ sudo samba-tool user create student1 Password@123
User 'student1' added successfully
sasmi@dc1:~$ samba-tool ou create "OU=Admins,DC=it83,DC=local"
```

```
sasmi@dc1:~$ sudo samba-tool user list
Guest
admin1
krbtgt
staff1
student1
Administrator
sasmi@dc1:~$
```

Part E – Web Server

The Apache HTTP Server Active

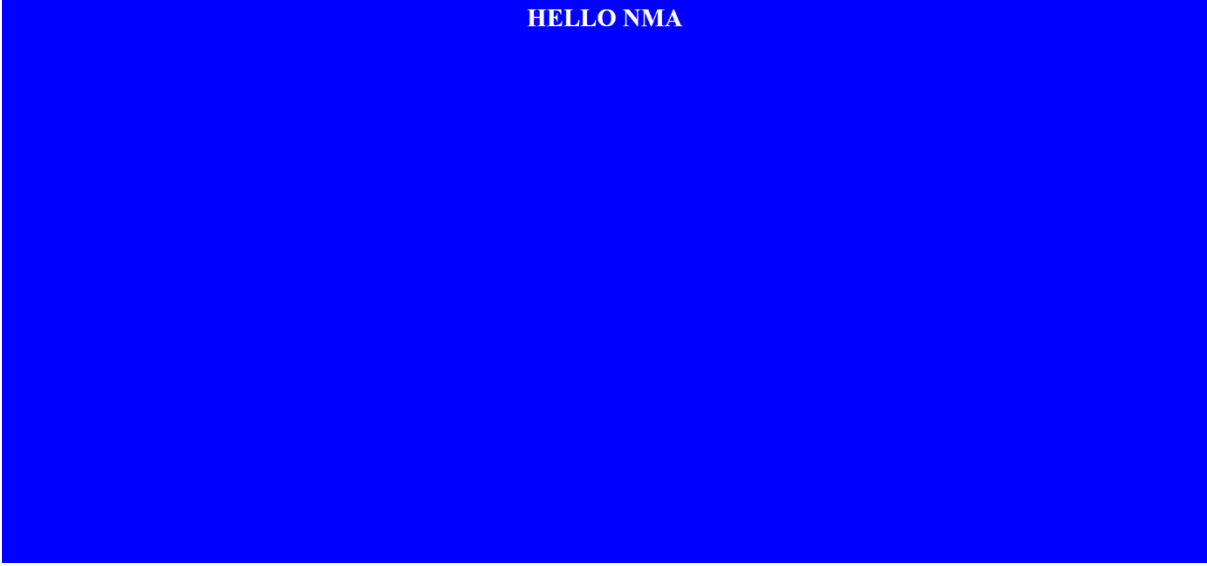
```
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset>
   Active: active (running) since Sun 2026-04-12 14:06:16 +0530; 23s ago
  Invocation: 887835767e724cf9b176787985110d88
     Docs: man:httpd.service(8)
  Main PID: 70655 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; B>
      Tasks: 177 (limit: 23100)
     Memory: 13.8M (peak: 14.3M)
        CPU: 344ms
    CGroup: /system.slice/httpd.service
            └─70655 /usr/sbin/httpd -DFOREGROUND
              └─70656 /usr/sbin/httpd -DFOREGROUND
                └─70657 /usr/sbin/httpd -DFOREGROUND
                  └─70658 /usr/sbin/httpd -DFOREGROUND
                    └─70659 /usr/sbin/httpd -DFOREGROUND

Apr 12 14:06:15 localhost.localdomain systemd[1]: Starting httpd.service - >
Apr 12 14:06:16 localhost.localdomain (httpd)[70655]: httpd.service: Refere>
Apr 12 14:06:16 localhost.localdomain httpd[70655]: AH00558: httpd: Could n>
Apr 12 14:06:16 localhost.localdomain systemd[1]: Started httpd.service - T>
Apr 12 14:06:16 localhost.localdomain httpd[70655]: Server configured, list>
lines 1-22/22 (END)
```

This figure confirms that the Apache service was installed and running on the Ubuntu server. It verifies that the server platform required for web publication was active and available to serve HTTP requests. This is important because the web page requirement cannot be met unless the web server process is correctly installed and operational

```
<!DOCTYPE html>
<html>
<head>
  <title>HELLO NMA</title>
  <style>
    body {
      background-color: blue;
      color: white;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
      font-family: Arial, sans-serif;
      font-size: 48px;
      font-weight: bold;
    }
  </style>
</head>
<body>
  HELLO NMA
</body>
</html>
```

This figure illustrates the source code used to create the assignment web page. It confirms that the page was properly configured to display the exact text HELLO NMA with the required blue background styling. This is important as the assignment evaluates not only whether the web server is running, but also whether the page content and design meet the specified requirements.



HELLO NMA

This figure provides the client-side validation of the published web service. It shows that the required HELLO NMA page was reachable from the client browser and displayed with the expected visual format. This is important because it confirms that the service was not only hosted locally on the server, but was accessible across the laboratory network as required by the assignment.